

AI-Based Tender Evaluation & Eligibility Analysis

for Government Procurement by CRPF

Round 1 Proposal | Decision-Support Platform for Formal Government Procurement

A platform that extracts eligibility criteria from tender documents, parses heterogeneous bidder submissions, produces explainable criterion-level verdicts, and preserves a courtroom-grade audit trail.

SUBMITTED BY

Team Conviction

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1 Problem Understanding — Beyond the Surface

The problem statement frames this as “extract criteria, parse bids, match, report.” That is the visible problem. To deliver a system a CRPF procurement officer can actually sign off on, we must respect what Indian government procurement really is.

1.1 The real operating context

- CRPF procurements are governed by the **General Financial Rules (GFR) 2017**, the **Manual for Procurement of Goods / Works / Services (2017)**, and **CVC guidelines**. Any AI-produced artefact that an officer signs must be defensible under all of these.
- Procurement decisions are **appealable**. A rejected bidder can file a representation or approach a court. Every verdict the system produces must survive adversarial scrutiny by someone whose business depends on proving it wrong.
- **RTI (Right to Information) applies**. Reports may be disclosed publicly. They must be clean of internal scratch-work and must not leak other bidders’ commercial data inappropriately.
- **Data sensitivity is high**. CRPF is a Central Armed Police Force. Some tenders touch restricted items. The architecture must support a **sovereign, on-premise** deployment — third-party cloud LLMs cannot be the default for sensitive tenders.
- Bidders are **sophisticated actors**. They present documents to maximise their chance of qualification. Turnover of **Rs.4.9 crore** may be rounded to “approximately **Rs.5 crore**” in a covering letter. The system must be robust to this without being punitive.
- **The officer is accountable — not the AI**. The system is a decision-support tool. It must make the officer faster and more consistent without ever taking her authority away.

i Key Insight

Most hackathon solutions will treat this as a document-Q&A problem. It is not. It is an *explainable, auditable, adversarial-robust decision-support problem inside a regulated procurement workflow*. Getting this framing right is the single biggest predictor of whether the downstream design is defensible.

2 North Star — What the Solution Must Be

We design every component of the system against six commitments:

1. Reduce evaluation time from **days to hours**, without reducing rigour.
2. Produce **identical verdicts** when the same bid is evaluated twice on the same system version.
3. **Zero silent rejections** — every borderline case is escalated, never buried.
4. Every verdict traces to: *which criterion, which document, which page, which value, which rule, which confidence*.
5. Work on **scanned, photographed, handwritten, bilingual, and stamped** Indian documents — not just clean PDFs.
6. Be **auditable years later** — the full decision trail must be reproducible bit-for-bit.

3 System Architecture — Layered View

The system is organised as seven layers, each with a clear responsibility and a clean contract with its neighbours. External verification services and the immutable audit ledger interact with multiple layers.

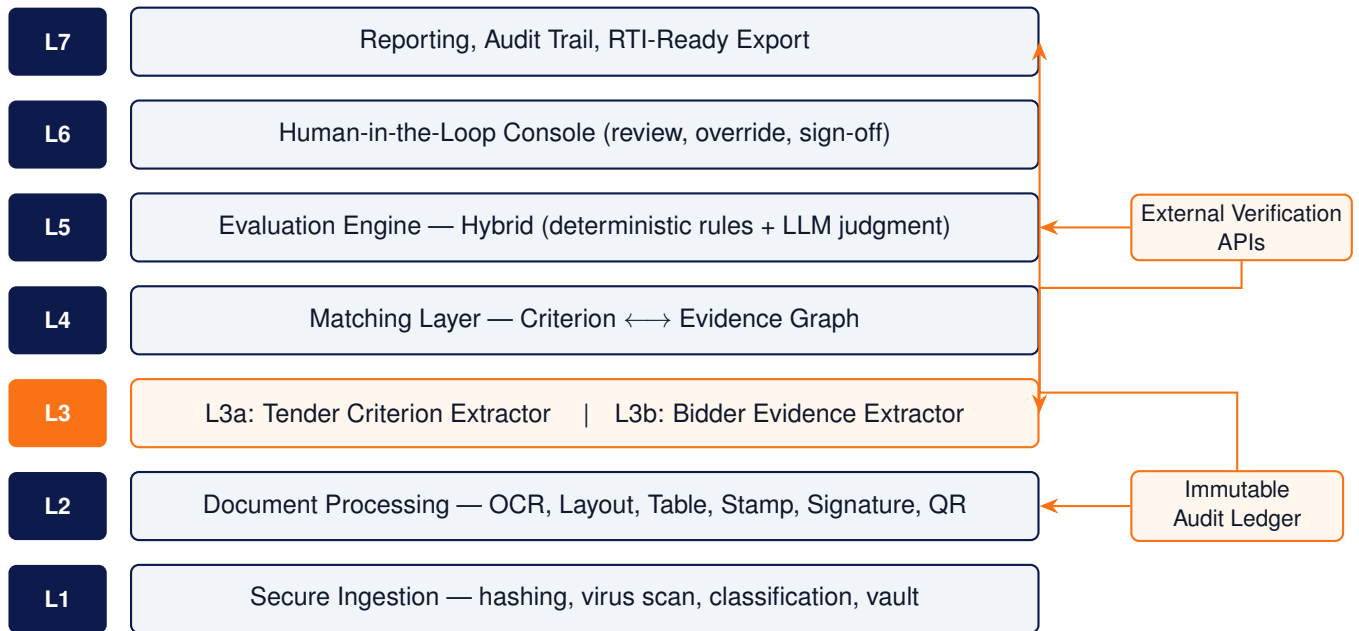


Figure 1: Seven-layer architecture with external verification services and the audit ledger as cross-cutting concerns. Layer 3 (orange) is the extraction core where the most domain-specific intelligence lives.

* Differentiator

The cross-cutting *Audit Ledger* (hash-chained, append-only) touches every layer. This is not a feature — it is an architectural decision. It is what makes every verdict the system ever produces reproducible, challengeable, and defensible under CVC audit or judicial review.

4 Layer 1 — Secure Ingestion

Every document that enters the system is:

- 1. Virus-scanned** using ClamAV or equivalent before any other processing touches it.
- 2. SHA-256 hashed** immediately. The hash is appended to the audit ledger. This fingerprint is the identity the rest of the system uses. If the same document is re-uploaded later, it is recognised. If a document is altered post-submission, the hash mismatch is flagged.
- 3. Classified** — tender vs bid, bidder identity, submission timestamp (from e-procurement portal metadata where available).
- 4. Sealed** in a per-tender evidence vault with role-based access control. Evaluators see only what the workflow grants them.

i Key Insight

This layer alone eliminates an entire class of disputes: “my document was altered after submission.” Hashes on file, timestamped, audit-logged — the claim is verifiable in seconds.

5 Layer 2 — Document Processing

A single Indian bid bundle can contain clean PDFs, 150-DPI skewed scans, mobile photographs with glare, Word files with embedded signature images, Excel sheets with merged cells, Hindi or regional-language certificates, and notarised ink-stamped paper. Our pipeline is designed for this reality.

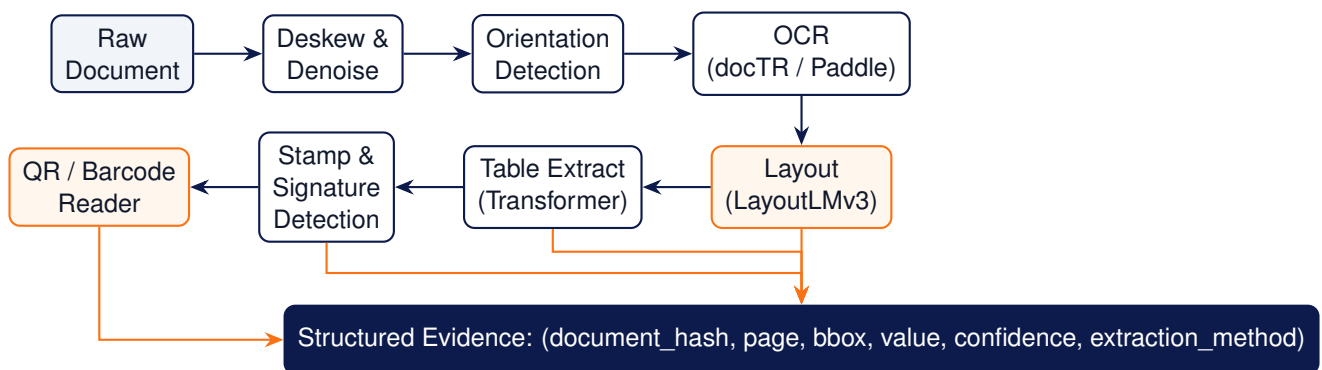


Figure 2: Document processing pipeline. Every extracted value carries provenance — document hash, page, bounding box, extraction method, confidence score.

5.1 OCR and layout stack

- **Primary OCR:** docTR or PaddleOCR — open-source, on-prem deployable, strong on Indian scans.
- **Cloud fallback (non-sensitive tenders only):** Amazon Textract, Google Document AI, Azure Document Intelligence.
- **Vernacular:** Tesseract with Devanagari and regional script packs; IndicOCR research models for complex scripts; translation cross-check for bilingual documents.
- **Layout-aware extraction:** LayoutLMv3 or Donut for semi-structured certificates, forms, and multi-column tender pages.

5.2 Specialised extractors

- **Tables:** Camelot + TableTransformer — dual-path, used heavily on balance sheets and turnover statements.
- **Signature and stamp detection:** YOLO-trained head that asks “is there a stamp/signature in this region?” — essential because many Indian documents are only valid when stamped and signed.
- **QR / barcode reader:** CA-issued certificates carry UDIN QRs; DigiLocker documents carry verifiable QRs; GST certificates carry QRs resolving to the GSTN portal. We scan these and verify at source with the issuing authority.

Every extracted value carries (document_hash, page_number, bounding_box, extraction_method, confidence). This tuple is the atomic unit of evidence for the rest of the system.

6 Layer 3a — Tender Criterion Extraction

The tender document is the source of truth. Getting this layer wrong poisons everything downstream, so we do it carefully and with an explicit officer-in-the-loop checkpoint before any bid is evaluated.

6.1 The Criterion DNA schema

Every criterion is cast into a structured schema that is explicit, machine-checkable, and reviewable.

```
{
  "criterion_id": "C-003",
  "source_clause_ref": "Section 5.2, clause (iv), page 14",
  "source_text_verbatim": "...minimum annual turnover of Rs 5 crore
    in at least 2 of the last 3 financial years...",
  "category": "financial",
  "mandatory": true,
  "type": "numeric_threshold",
  "parameters": {
    "metric": "annual_turnover",
    "operator": ">=",
    "value": 50000000,
    "currency": "INR",
    "window": { "years_back": 3, "min_years_satisfying": 2 }
  },
  "acceptable_evidence": [
    "audited_balance_sheet",
    "ca_certificate_turnover",
    "income_tax_return_with_computation"
  ],
  "evidence_weight": {
    "ca_certificate_turnover": 1.0,
    "audited_balance_sheet": 1.0,
    "self_declaration": 0.3
  },
  "extracted_by": "llm_with_rule_check",
  "extraction_confidence": 0.94
}
```

Figure 3: The *Criterion DNA* schema. Every tender clause is structured into a form the evaluation engine can deterministically check.

6.2 Criterion type taxonomy

This taxonomy is what makes the downstream evaluation engine deterministic where it can be, and fuzzy only where it has to be.

Type	Example
numeric_threshold	Minimum turnover of Rs.5 crore
count_threshold	At least 3 similar projects in the last 5 years
binary_compliance	Valid GST registration
date_validity	ISO 9001 valid on tender opening date
document_presence	Power of Attorney enclosed
categorical_match	Class-I local supplier as per PPP-MII 2017
semantic_similarity	Projects of “similar nature and magnitude”
composite	Boolean combinations of the above

Table 1: Criterion type taxonomy. Types 1–6 are deterministic rules; type 7 is LLM-assisted; type 8 combines them.

6.3 Mandatory vs optional detection

Detected through (a) explicit language cues — *shall, must, mandatory* vs *desirable, preferred, may* — and (b) structural position — clauses under “Eligibility” default to mandatory; clauses under “Evaluation / Marking Scheme” are usually scored. The classification is always surfaced to the officer for review before bids are evaluated.

6.4 Corrigendum handling

Tenders are routinely amended. Every corrigendum is ingested, diffed against the base tender, and merged into the live criterion set with full version history preserved. The officer sees, e.g., “*Criterion C-003 amended by Corrigendum 2 dated 14-Nov-2026.*”

i Key Insight

No criterion is ever locked in by the machine alone. Before any bid is evaluated, the extracted criterion list is presented to the procurement officer for approval. This single control prevents the system from silently misreading the tender itself.

7 Layer 3b — Bidder Evidence Extraction

Extraction is **criterion-driven**, not document-driven. For each bidder, we do not try to pull everything; we pull exactly what the criterion list demands. This keeps extraction targeted, performant, and auditable.

Document-greedy extractors — the common default — try to pull every field from every document and match to criteria later. This scales poorly and produces noise that clouds the evaluator’s view. By contrast, our approach treats each criterion as a *query* and the bidder’s document set as the *evidence pool*; extraction happens only in service of answering those queries.

7.1 The bidder document reality

A typical Indian bid bundle spans several formats, each with its own failure modes. Our extractor applies a format-specific strategy while producing a uniform downstream evidence schema.

Document Type	Typical Challenge	Our Handling
Typed digital PDFs	Clean text but inconsistent layout across bidders	LayoutLMv3 with structural anchors
Scanned PDFs (low DPI)	Skew, noise, bleed-through from reverse page	Deskew + denoise + dual-OCR voting
Mobile photographs	Glare, shadow, curvature, fingers in frame	Perspective correction, local contrast normalisation
Excel / spreadsheets	Merged cells, split headers, unit mismatches	SheetJS + header-region classifier + unit normaliser
Vernacular certificates	Mixed-script, Devanagari stamps, regional fonts	Indic OCR + translation cross-check
Notarised / stamped paper	Ink bleed, overlapping stamps on text	Stamp segmentation + text in-painting before OCR
QR-bearing documents	QR decoding plus live verification at issuer	Native QR decoder + direct source verification

Table 2: Indian bidder document types and our per-format handling. All paths converge on the same downstream Evidence Object schema.

i Key Insight

Criterion-driven extraction has an under-appreciated audit benefit: the system cannot “invent” a finding the officer did not ask it to look for. Every extracted value exists only because some criterion demanded it — and that criterion traces straight back to a specific clause of the tender.

i Key Insight

Criterion-driven extraction has an under-appreciated audit benefit: the system cannot “invent” a finding the officer did not ask it to look for. Every extracted value exists only because some criterion demanded it — and that criterion traces straight back to a specific clause of the tender.

7.2 Dual-path extraction — deliberate redundancy

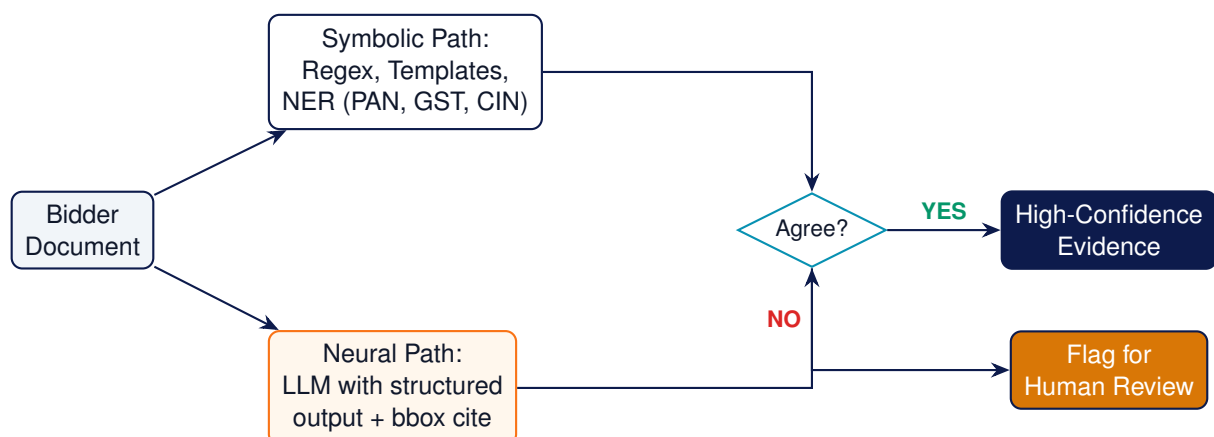


Figure 4: Dual-path extraction: symbolic (regex/template) and neural (LLM) paths run independently. Agreement drives confidence; disagreement forces human review.

* Differentiator

We never route solely on a single extraction path. The dual-path discipline dramatically reduces silent errors — if the regex and the LLM disagree, the bidder is *not* penalised; the case is escalated with both candidate interpretations shown.

7.3 The evidence object

Each extracted value becomes an Evidence Object bound to one or more criteria, with full provenance and (where applicable) external verification results:

```
Criterion C-003 <-> Evidence E-017
source:      ABCCo_balance_sheet.pdf   (hash: 9f2a7c...)
page:        6
bbox:        [122, 410, 478, 452]
value:       { fy: "2023-24", turnover_inr: 72500000 }
extraction:  "llm+regex"
confidence:  0.97
verification: {
  ca_udin_check: "PASSED (UDIN 24XXXXX verified at ICAI)",
  cross_check_ITR: "MATCH"
}
```

Figure 5: An evidence object carries full provenance plus external-source verification results.

8 Layer 4 — Matching: The Evidence Graph

Real bids don't match one-to-one. Turnover may be evidenced by a CA certificate *and* a balance sheet *and* an ITR (three pieces of evidence for one criterion). A single document — say a company profile — may touch five criteria. We model this as a bipartite, weighted **Evidence Graph**. Every verdict is literally a sub-graph an officer can visualise.

8.1 Why a bipartite weighted graph

A flat matching table silently assumes each piece of evidence contributes equally and independently. That assumption breaks on real Indian bids in three distinct ways:

- **Many-to-one.** A turnover criterion may accept a CA certificate, a balance sheet, and an ITR — three pieces of evidence for one criterion, any of which may be individually sufficient.
- **One-to-many.** A company-profile brochure may carry registered address, CIN, GST number, and an experience summary — one document touching four criteria simultaneously.
- **Partial contribution.** A self-declaration alone may contribute only a weight of 0.3 toward a criterion, but paired with a CA certificate the combined weight crosses the satisfaction threshold.

A bipartite weighted graph encodes all three relationships cleanly. The officer can inspect any verdict as a sub-graph rather than trusting a hidden scoring formula.

A criterion is satisfied when the sum of incident edge weights reaches 1.0. A sum in [0.5, 1.0) routes to *Needs Review*; below 0.5 the criterion fails. These thresholds are configurable, logged in the audit trail, and frozen per tender once the officer approves the criterion set.

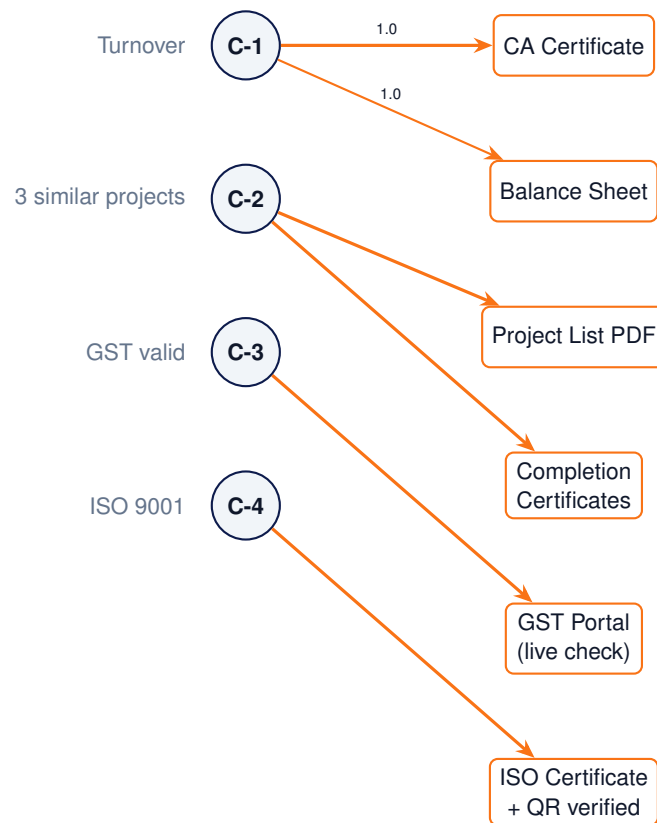


Figure 6: Evidence Graph (snippet for one bidder). Criteria on the left, evidence on the right, weighted edges. Every verdict is a traversal of this graph the officer can inspect.

8.2 Semantic similarity — the hardest case

“At least 3 similar projects in the last 5 years” is not a number match. We decompose it into measurable sub-factors:

- **Semantic embedding similarity** of project descriptions (tuned domain embeddings).
- **Value band** within tender-specified range, e.g. $\pm 50\%$ of estimated contract value.
- **Duration band.**
- **Nature** — construction vs consulting vs supply (hard category check).
- **Client type** — government, PSU, private, where the tender restricts it.

A project with 0.88 semantic similarity but in the wrong value band does **not** silently pass. The similarity score is surfaced with its contributing factors and the officer decides.

9 Layer 5 — Evaluation Engine (Hybrid by Design)

Most hackathon solutions will use one big LLM prompt. That is the wrong answer for government work — unauditable, non-deterministic, brittle under challenge.

Our engine is **hybrid**:

- **Deterministic rules engine** for well-defined criterion types (numeric, count, date, binary, document presence, categorical, composite). Same input $\Rightarrow$ same verdict, every time, forever. These cover roughly 80% of real criteria.

- **LLM-backed judgment** only for the genuinely fuzzy cases — semantic similarity, cross-document reasoning, ambiguous tender language. Even here, the LLM returns a structured score with a rationale, not free-form prose.

9.1 Confidence-gated routing

Every verdict carries a **calibrated** confidence. Calibration via isotonic regression on a held-out labelled set — so that “0.9 confidence” actually corresponds to 90% historical accuracy. Uncalibrated model scores are systematically over-confident; we refuse to ship that.

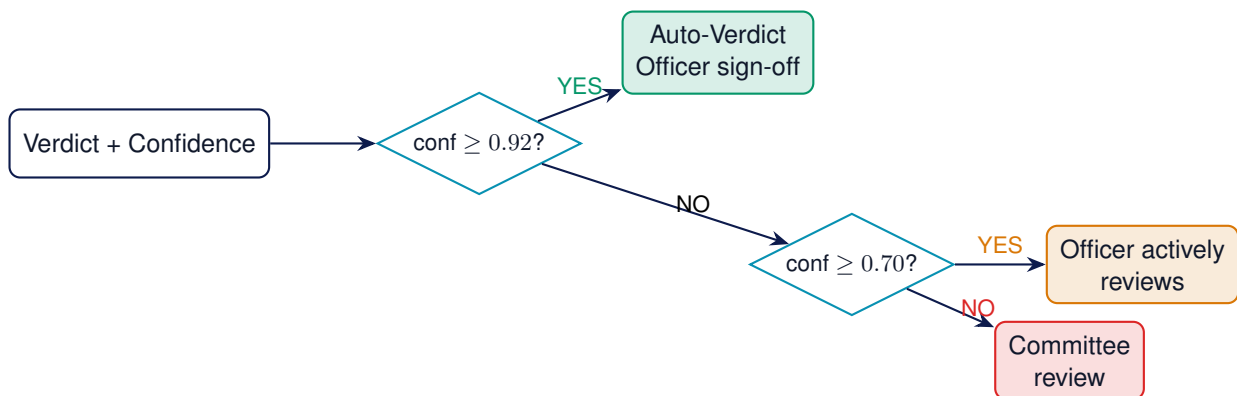


Figure 7: Confidence-gated routing. Thresholds are tunable per procurement authority.

9.2 Overall bidder verdict

A composite of criterion-level verdicts under the following boolean rule:

- Any **mandatory FAIL** $\Rightarrow$ bidder FAIL.
- Any **mandatory NEEDS_REVIEW** with all others PASS $\Rightarrow$ bidder NEEDS_REVIEW.
- All mandatory PASS $\Rightarrow$ bidder PASS.

10 Layer 6 — Human-in-the-Loop Console

The console is designed so the officer is always in authority, never overwhelmed.

10.1 Design commitments

- The officer never sees raw model output. She sees **criterion, verdict, confidence, supporting evidence with page preview and highlighted bounding box, rationale**.
- One-click drill-down to the evidence graph sub-tree for any verdict.
- Inline override with a **mandatory reason string**. Overrides are logged as officer-actions, distinct from system-verdicts.
- Side-by-side comparison grid for the evaluation committee.
- Automated **clarification letter drafts** for bidders when a document is ambiguous or incomplete — saves hours of manual drafting.
- Officer feedback on resolved NEEDS_REVIEW cases feeds the active-learning loop that improves routing over time.

10.2 UI mockup — criterion-level evaluation view

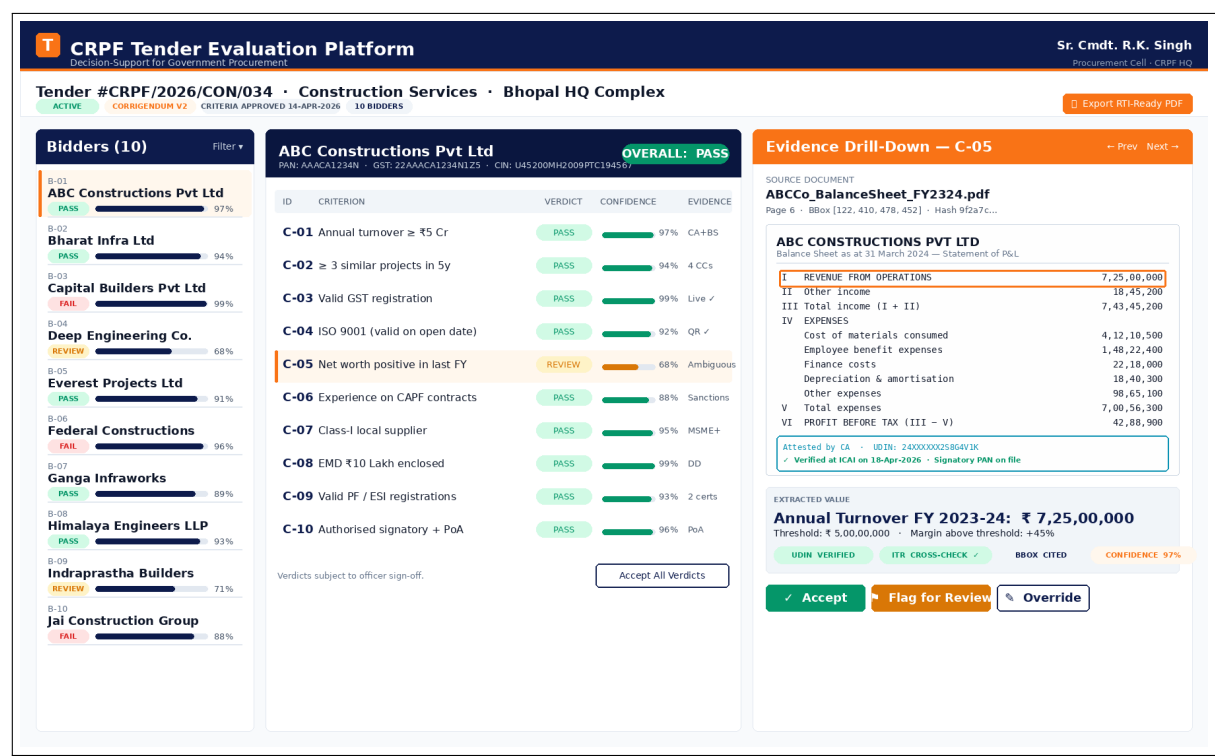


Figure 8: Criterion-level evaluation view.

10.3 UI mockup — committee side-by-side grid

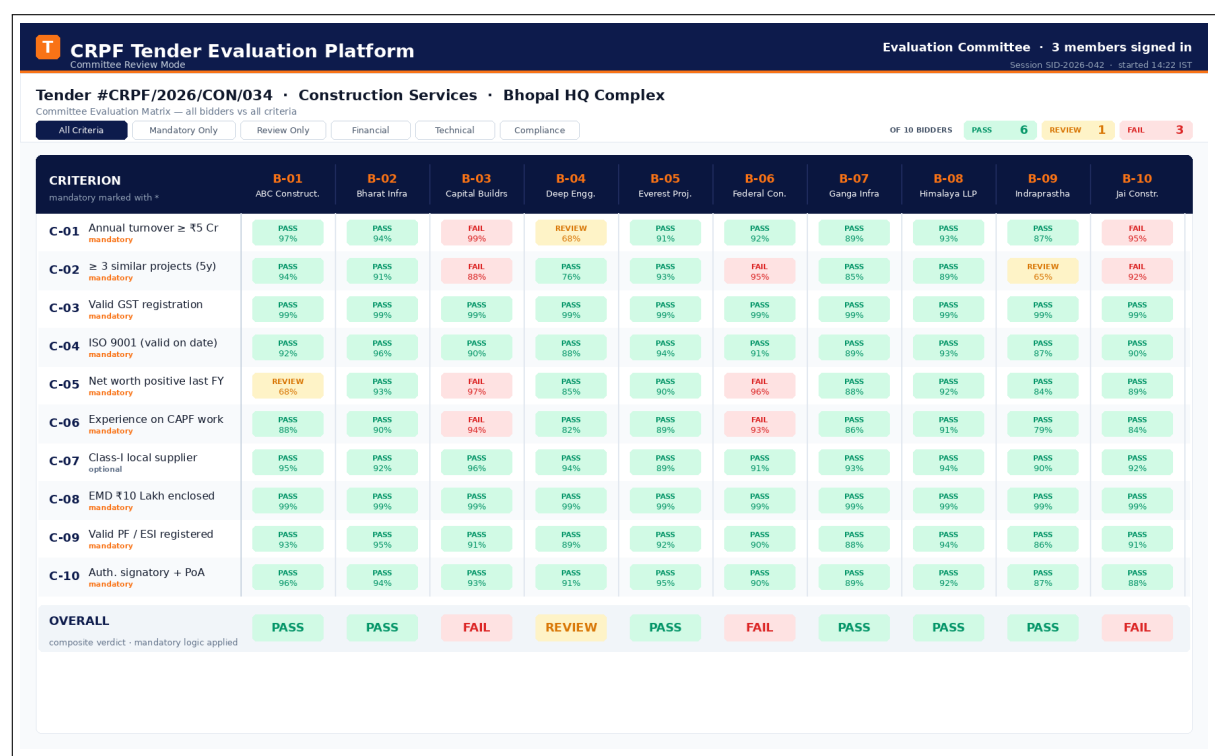


Figure 9: Committee side-by-side grid.

11 Out-of-the-Box Differentiators

This section is what separates a winning proposal from a competent one. Each item is concrete, implementable, and tied to a real procurement pain point.

11.1 Live government API verification

A GST number is only meaningful if currently active; a PAN must resolve to the claimed entity. We integrate:

- **GSTN API** — GSTIN active status, registered entity name.
- **MCA21** — CIN validity, company status (active / strike-off / under liquidation), directors.
- **PAN verification** (NSDL / UTIITSL).
- **EPFO / ESIC** — establishment-code validity (required in labour-heavy tenders).
- **DigiLocker** — if the bidder submitted a DigiLocker-issued document, verify the URI at the issuing authority.
- **UDIN verification** (ICAI) — every CA-attested document carries a UDIN. Verifying it at ICAI kills most forged CA certificates. **Almost no hackathon team will know about this.**

11.2 Debarred / blacklisted vendor screening

Automatic check against the CVC debarred list, World Bank debarred list, state-level blacklists, and CRPF’s own historical blacklist. A blacklisted bidder is flagged immediately, regardless of other criteria.

11.3 Document authenticity and forgery detection

- **Error-Level Analysis (ELA)** on scanned certificates — reveals tampered regions.
- **Font consistency** — a single figure in a slightly different font is suspect.
- **Noise / compression artefact analysis** — regions re-saved after edits leave JPEG signatures that differ from their surroundings.
- **Metadata inspection** — a “2023 certificate” with EXIF creation date of last week is a red flag.
- **Stamp and signature consistency** — the same authorised signatory’s signature across bidder documents should look identical.

None of these are conclusive alone. Together they produce a **Document Authenticity Score** surfaced to the officer.

11.4 Cross-bidder cartel / collusion detection

When evaluating multiple bids in parallel, the system looks for suspicious patterns:

- Common directors across bidders (via MCA).
- Shared registered addresses.
- Near-identical document formatting, fonts, or typos (classic shared-template tell).
- Sequential CA-UDIN numbers issued back-to-back to multiple bidders.
- Metadata overlap — identical MS Word “Author” fields across bids.

These do not disqualify. They raise a **Collusion Risk Flag** for the committee — exactly what CVC audits look for.

11.5 Temporal / validity checking

Every certificate is auto-checked for validity **on the tender opening date** — not the submission date, not today. Expired GST, lapsed ISO, out-of-date factory licences — caught automatically.

11.6 Vernacular and bilingual handling

Hindi-English bilingual documents are common. We support Devanagari OCR, regional script OCR, IndicBERT / MuRIL embeddings for vernacular semantic understanding, and Devanagari stamp/seal OCR.

11.7 Stamp, seal, and signature verification

Authorised-signatory matching (the bid covering letter's signature vs the board resolution's), mandatory-stamp presence checks, and notary / gazette attestation detection.

11.8 Evidence weight model

Not all evidence is equal. Self-declaration < CA-certified < government-issued. The weight model is explicit, tunable per tender, and visible in the audit trail.

11.9 Ambiguous tender criteria flagging

A genuinely novel capability: the system flags ambiguous tender criteria *to the officer before bids come in*. "Projects of similar nature" with no further definition triggers a prompt to add a working definition, which then governs evaluation. Prevents post-hoc disputes.

11.10 Cross-criterion consistency checks

If a bidder claims **Rs.50** crore turnover in the covering letter but the balance sheet shows **Rs.5** crore, we catch that within-bid inconsistency even if each value individually passes a criterion.

11.11 Cryptographic audit trail

Every document hash, extraction, rule firing, verdict, and officer override is appended to a hash-chained ledger. Each entry references the SHA-256 of the previous. You cannot rewrite history without the chain breaking. This is the single feature that makes the system defensible years later.

11.12 RTI-ready export

One-button generation of an RTI-disclosable report per bidder — redacted of other bidders' commercial info, clean of internal scratch-work, complete audit trail.

11.13 Fairness auditing

The system self-checks whether it systematically under-verdicts bids from MSEs, startups, or specific regions. GFR explicitly supports MSE/startup preference; the system should support, not obstruct it.

11.14 Replay mode

Any past evaluation can be replayed — load historical hashes, pinned criterion version, pinned system version, and reproduce the exact verdicts. Gold standard for auditability.

* Differentiator

If you highlight only three differentiators in a presentation, make them: **(1) UDIN verification at ICAI** — kills forged CA certificates at source, **(2) cross-bidder collusion detection** — the kind of thinking a procurement veteran does, not just an engineer, and **(3) flagging ambiguous criteria before bids arrive** — flips the problem on its head and prevents disputes.

12 Technology Choices (and Why)

Layer	Primary Choice	Why
OCR	docTR / PaddleOCR	Open-source, on-prem, strong on Indian scans, CPU-friendly
Layout	LayoutLMv3	Best-in-class for form + table-heavy documents
Tables	TableTransformer + Camelot	Dual-path, robust on scanned tables
NER / pattern	spaCy + custom regex	Deterministic and auditable for Indian IDs
LLM (sovereign)	Llama 3 / Mistral on-prem; Sarvam / Krutrim for Indic	No data leaves the perimeter
LLM (cloud-allowed)	Claude / GPT-4 via gateway	Quality — only for non-sensitive tenders
Embeddings	BGE-M3 / MuRIL	Multilingual, Indian-language aware
Vector store	Qdrant / Milvus (self-hosted)	On-prem, mature
Rules engine	Drools or lightweight Python rules	Deterministic, testable, auditable
Orchestration	Temporal / Prefect	Durable workflows, retrievable steps, visible state
Audit ledger	PostgreSQL append-only + hash chain	Simple, provable, no blockchain theatre
Deployment	Docker + Kubernetes on-prem, STQC-certified DC	Meets CRPF security posture

Table 3: Technology stack with deliberate choices for sovereignty, auditability, and determinism.

12.1 Deliberate non-choices

- We do not rely on a single LLM prompt for final verdicts.
- We do not use external cloud LLMs in the default path for sensitive CRPF tenders.
- We do not build on closed / proprietary document-AI that can disappear behind a paywall or

a pricing change.

13 Risks and Trade-offs — Honest View

Risk	Impact	Mitigation
OCR errors on poor scans	Wrong values extracted	Dual-path extraction, low-confidence routing to review, never silent
LLM hallucination	Fabricated evidence citation	Structured outputs require a bbox citation — LLM outputs without it are rejected
Mis-read mandatory vs optional	Wrong disqualification	Officer must approve criterion list before bids are evaluated
Adversarial bidders	Documents crafted to fool the system	Dual-path, forgery detection, live API verification, weighted evidence
Model drift over time	Verdicts change as models update	Frozen model versions per tender; replay mode pins version
Vernacular edge cases	Misreading regional certificates	Explicit NEEDS_REVIEW for low-confidence vernacular; officer escalation
Over-automation	Officer rubber-stamps	UI forces active review on NEEDS_REVIEW; sign-off is a deliberate, reasoned action
Data sensitivity leak	Leak of bid info	On-prem default, RBAC, audit on every read, no external API for sensitive tenders
“AI did it” legal defence does not hold	Officer must own the decision	System is framed as decision-support, not decision-maker; every report ends with “Verdicts subject to officer sign-off”

Table 4: Primary risks and our mitigations. The design consistently chooses paranoia over speed.

! Risk

The single biggest trade-off is **speed vs paranoia**. We consistently choose paranoia. A faster system that silently mis-qualifies a bidder is, in this domain, strictly worse than a slower system that escalates. Government procurement is an error-averse context; we design for that, not against it.

14 Round 2 Implementation Plan (Sandbox)

Assuming a sandbox with representative mock tender and bidder documents.

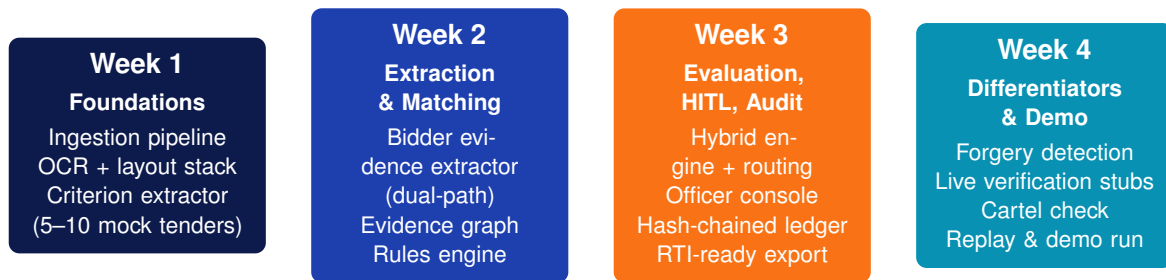


Figure 10: Four-week sandbox implementation plan. Ends with a live demo on a 10-bidder mock tender, producing a consolidated report with verdict breakdown, full evidence trail, visible audit ledger, and one-click RTI export.

Demo deliverable: Upload a mock tender and 10 mock bids → the system extracts criteria → the officer reviews and approves → bids are processed → a consolidated report shows, e.g., 6 PASS, 3 FAIL, 1 NEEDS_REVIEW, each with full evidence, visible audit ledger, and RTI-ready export generated in a single click.

15 Closing Note — What We Are Really Building

The narrow framing of this problem is “extract, match, report.” The real opportunity is to build the **first serious decision-support platform for Indian government procurement** — one an officer can defend under any challenge, a bidder can trust to be fair, CVC can audit on demand, and that makes the entire process faster *and* cleaner at the same time.

Every design decision in this document serves that. No silent rejections. No opaque LLM verdicts. No broken audit trails. No ignored vernacular documents. No missed expired certificates. No undetected cartels.

That is the system we will build.